HW7

Math 742

2026-04-07

Problem 1: Bootstrap, Jackknife, and Bias Correction

Let $T_n = T(X_1, \ldots, X_n)$ be an estimator of a parameter θ based on an i.i.d. sample.

(a) Define the **bootstrap estimate of bias** of T_n.

(b) Define the **jackknife estimate of bias** of T_n.

(c) Explain the main idea behind **bias correction** using the bootstrap.

(d) Why does the jackknife tend to work better for smooth statistics such as the sample mean than for non-smooth statistics such as the sample median?

Problem 2: Bayesian Foundations: Prior, Likelihood, Posterior

Suppose $X \sim \text{Binomial}(n, \theta)$ and that $X = x$ is observed.

(a) Write down the likelihood function $L(\theta \mid x)$.

(b) In words, explain the roles of:

- the prior
- the likelihood
- the posterior

(c) State Bayes' rule in this setting, writing the posterior in proportional form.

(d) Explain what it means to say that the posterior is a "compromise" between the prior and the likelihood.

(e) Under what general circumstance will the likelihood dominate the prior?

Problem 3: Conjugate Priors: Beta–Binomial and Normal–Normal

(a) Beta-Binomial

Suppose

$$X \mid \theta \sim \text{Binomial}(n, \theta), \qquad \theta \sim \text{Beta}(\alpha, \beta).$$

Derive the posterior distribution of $\theta \mid X = x$ and identify its parameters.

(b) Explain why α and β are often interpreted as prior successes and prior failures.

(c) Normal-Normal

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2), \qquad \sigma^2 \text{ known,}$$

and the prior is

$$\mu \sim N(\mu_0, \tau^2).$$

Derive the posterior mean of μ.

(d) Show that the posterior mean can be written as

$$E(\mu \mid x) = w\bar{X} + (1-w)\mu_0$$

for some weight w, and give the formula for w.

(e) Explain why this posterior mean is called a **shrinkage estimator**.

Problem 4: Model Selection: AIC, BIC, and the Likelihood Principle

Suppose two models are fit to the same dataset. Model M_1 has k_1 parameters and maximized log-likelihood ℓ_1, while model M_2 has k_2 parameters and maximized log-likelihood ℓ_2.

(a) Write down the formulas for AIC and BIC.

(b) Explain in words what each criterion is trying to balance.

(c) Why does BIC usually penalize model complexity more strongly than AIC?

(d) State the likelihood principle in words.

(e) Briefly discuss whether AIC and BIC are based only on the likelihood, or whether they also incorporate additional considerations.

Problem 5: Penalization and Regularization: Ridge and Lasso

Consider the linear regression model

$$y = X\beta + \varepsilon.$$

(a) Write down the optimization problem for **ridge regression**.

(b) Write down the optimization problem for **lasso regression**.

(c) Explain the difference between an L_2 penalty and an L_1 penalty.

(d) Why does ridge regression usually shrink coefficients toward zero without setting many exactly equal to zero, while lasso can produce sparse solutions?

(e) Explain how regularization changes the bias–variance tradeoff.

(f) What prior distribution on the regression coefficients corresponds naturally to ridge regression? What prior corresponds naturally to lasso?